

Rethinking Digital Preservation

Applying the InterPlanetary File System for Sovereign and Resilient Archiving

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About me

- Research focus: decentralized, sovereign science infrastructure
- Relevant publications:
 - Incentive mechanisms in P2P networks (ACM Comput. Surv. 2023)
 - NAT traversal in IPFS — large-scale measurement campaign (2025)
 - *Academic Storage Cluster* (JCDL 2021) — early precursor
- Upstream contributor to **IPFS Kubo** (on-demand pinning, [PR #11252](#))
- Today: **D-LOCKSS** — a decentralized archiving initiative built on IPFS

What this talk is about

1. Why long-term preservation of the scholarly record is **at risk**
2. Why existing systems like Portico and LOCKSS are **not enough on their own**
3. How **IPFS** and modern P2P building blocks can help
4. **D-LOCKSS** — a decentralized archiving network
5. What this means for **libraries, archives, and researchers**

The scholarly record is disappearing

Eve (2024)⁴ examined **7.4 million** articles with a DOI:

- More than **2 million** are not preserved in **any** digital archive.

- Fewer than **half** are preserved in **more than one**.

These are measured numbers, not projections.

⁴ Eve, M. P. (2024). Digital Scholarly Journals Are Poorly Preserved: A Study of 7 Million Articles. *Journal of Librarianship and Scholarly Communication*, 12(1).

Open access is not "forever"

Laakso, Matthias, and Jahn (2021)⁵:

- **176 open-access journals** disappeared from the web between 2000 and 2019.
- Their articles disappeared with them.
- No archive, no backup, no access.

Scholarly memory erodes quietly, even when nobody is acting in bad faith.

⁵ Laakso, M., Matthias, L., & Jahn, N. (2021). Open Is Not Forever: A Study of Vanished Open Access Journals. *Journal of the Association for Information Science and Technology*, 72(9), 1099–1112.

Today's archives — partial solutions

Model	Example	Risk / Gap
Centralized	Portico	Single institutional custodian
Distributed, manual	LOCKSS / CLOCKSS	Polling overhead to prevent bit-rot, relies on URLs instead of content addressing
Content- addressed	IPFS	No persistence mechanism by itself
Token- incentivized	Filecoin	Token economy instead of mission

Portico — trusting a single custodian

- Publishers deposit content with **one** organization (ITHAKA).
- More than **1,000 libraries** rely on this single point of trust.
- If ITHAKA shuts down or changes policy: content is put at risk.

Centralized archives are **convenient** — but **structurally fragile**.

LOCKSS — "Lots of Copies Keep Stuff Safe"

Stanford, since 1999:

- Libraries operate their own **LOCKSS boxes**.
- Replication and bit-rot detection via **opinion polling** between peers.
- Established the principle of decentralized, cooperative preservation.

Two preservation models: Light vs. Dark

The LOCKSS technology is deployed in two distinct ways:

1. Local "Light Archives" (Standard LOCKSS)

- **Purpose:** Protects a specific library's active subscriptions.
- **Access:** Local backup. If the publisher goes offline, the library serves the content directly to its own patrons.

2. Global "Dark Archive" (CLOCKSS)

- **Purpose:** Ultimate fail-safe for the global scholarly record.
- **Access:** Content is locked away. Only made public (Open Access) globally if a "trigger event" occurs (e.g., publisher goes bankrupt).

The limits of opinion polling

But: the model comes from an era of **small, manually curated** networks.

- Each node requires **manual configuration** to track peers.
- The **opinion polling protocol** requires peers to constantly exchange votes to detect bit-rot (Today, this can be solved through self-verifying CIDs)
- Relies on **web crawling** with stable URLs (no content addressing).

LOCKSS established the principle. **We want a scalable and modern successor**

What we want from long-term preservation

1. **Integrity** — content unchanged, cryptographically verifiable
2. **Redundancy** — no dependence on any single institution
3. **Scalability** — from 12 nodes to hundreds or thousands
4. **Sovereignty** — no central control, no economical hurdles
5. **Low barrier to participation** — every GB counts

A closer look into content addressing

Today's web identifies files by **location** (URL). When the location dies, the file is gone with it.

Content addressing identifies a file by its **cryptographic hash**:

- Same file → same identifier, anywhere in the world.
- Any copy can be **verified bit-for-bit** against that identifier.
- Bit-rot becomes **detectable** instead of silent.
- Provenance and integrity stop depending on any single host.

For preservation that single change is decisive: the identity of an object stops depending on a particular URL or institution.

IPFS — content addressing in practice

Content addressing is a **principle**. **IPFS** is one tool that implements it well, and the one we build on:

- Files identified by **CID** (cryptographic content identifier)
- Peer-to-peer transport via **libp2p**
- Global discoverability via the **Distributed Hash Table** (DHT)
- Open standard with multiple implementations (Kubo, Helia, ...)

Why IPFS alone is not enough

- IPFS provides **no persistence guarantee**: unpinned content is garbage-collected.
- IPFS Cluster adds CRDT-based pinning — but only in a whitelisted limited group.

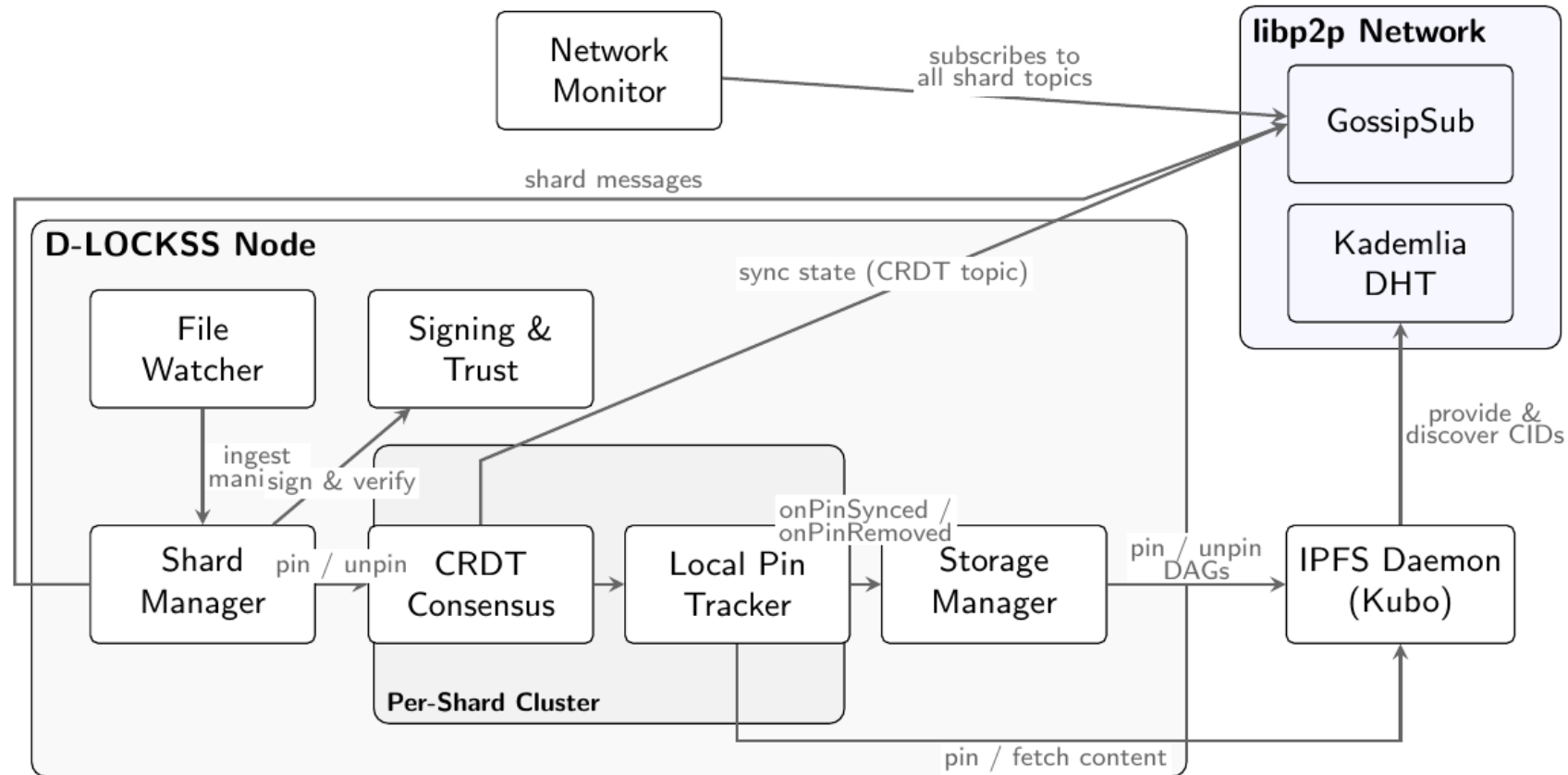
D-LOCKSS — design

We combine the preservation philosophy of **LOCKSS** with the scalability of **IPFS**.

Three core ideas:

1. **Signed Research Objects** — cryptographic provenance
2. **Per-shard CRDT replication** — automatic, without central coordination
3. **Dynamic sharding** — grows and shrinks with the network

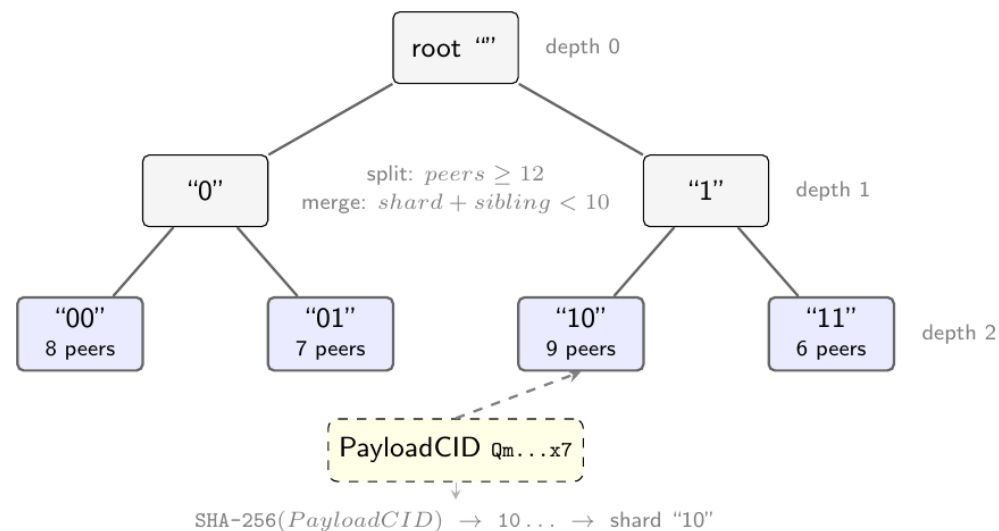
D-LOCKSS — node architecture



Dynamic sharding

Like a library that organizes books by call-number prefix and splits shelves once they get too full.

- Each node belongs to a **shard** identified by a binary prefix of its peer ID hash.
- Content is mapped to the matching shard via a hash of its CID.
- **Splits** above 12 peers, **merges** when sibling shards drop below 10.



Cross-shard with the "Tourist Pattern"

What if I ingest content that does *not* belong to my shard?

1. The node **subscribes briefly** to the target shard as an observer.
2. It pins the manifest there.
3. It publishes the ingest message.
4. It **leaves** the target shard again.

→ Any node can ingest any content. The network **routes automatically** to the responsible replication group.

On-Demand Pinning — preservation at the IPFS layer

A complementary contribution, **upstream to IPFS Kubo itself**:

github.com/ipfs/kubo/pull/11252

- A node **monitors** the DHT provider count for CIDs it cares about.
- If the count falls **below a target** (e.g., 5 providers): the node **automatically pins** the content.
- If the count stays well **above target**, it can **release** the pin again.
- **Storage-budget aware** — Respects existing IPFS storage limits.

Result: any IPFS node can become an **autonomous preservation rescuer** for content at risk — without committing to permanent storage of everything.

What this means in practice for libraries

- **Custody stays with the institutions.** No outsourcing to a single custodian; if a publisher disappears, copies persist verifiably elsewhere.
- **Low hardware barrier:** a single server, VM, or Raspberry-Pi.
- **Open source** (MIT/Apache-2.0), auditable, inspectable.
- **Configurable consortia** (open or allowlist-based).
- **No economic hurdles:** incentives come from mission and reciprocity, not from volatile storage markets.
- **FAIR + NDSA Level 4:** content becomes Findable (via CID), Accessible, Interoperable, Reusable, with automatic repair across geographically distributed institutions.

Status and roadmap

Today:

- D-LOCKSS prototype in Go, open source
- Testnet harness (`runtestnet.sh`), Docker-Compose deployment
- Live monitor dashboard (Wikimedia Cloud)
- On-demand pinning: upstream PR under maintainer review

Next steps:

- Multi-institutional pilot deployment
- Empirical evaluation (replication time, resilience, resource consumption)
- Search and semantic retrieval

Take-aways

1. The scholarly record is **already** insufficiently preserved, and the dominant archives are structurally fragile.
2. **Content addressing** is the property we need; **IPFS** is one tool that implements it well.
3. **D-LOCKSS** modernises "Lots of Copies Keep Stuff Safe".
4. **On-demand pinning** lets any IPFS node act as an autonomous preservation rescuer — contributed upstream to IPFS Kubo.
5. Sovereign archiving is operationally feasible today, and we are looking for partners to make it visible.

Thank you.

Be part of archiving the worlds knowledge.

Every node makes a difference.

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D-LOCKSS testnet: github.com/gipplab/D-LOCKSS

On-demand pinning: [ipfs/kubo](https://ipfs.io/ipfs/kubo) PR #11252

Track: Datensouveränität (Data Sovereignty) · Session: Digital Archiving